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	The National Research Council's Space Studies Board has stated that a principal physiologic hurdle to man's extended presence in space is the osteopenia and sarcopenia which parallels reduced gravity. The extent of the loss is extremely high, approaching a decrease in bone mineral density (BMD) in the lower appendicular skeleton at a rate of 1.6% per month and reducing maximal voluntary contractions of some muscle groups at a rate of 5% per month. Interestingly, the amount of bone (and muscle) loss between individual astronauts is highly variable with some astronauts losing large amounts of tissue while others are largely unaffected. This large individual variability, which has also been observed during bedrest and immobilization studies on Earth, may be accounted for, at least to a large extent, by genetic variations which may give rise to a differential mechanosensitivity of the musculoskeleton. We have collected preliminary data demonstrating that genetically distinct inbred strains of mice also demonstrate a distinct sensitivity to		

Task Description:	conditions of simulated weightlessness; while as much as 60% of trabecular bone is lost in the hindlimbs of BALB/cByJ mice within 3 weeks of disuse, the same conditions leave trabecular bone quantity and quality nearly unchanged in C3H/HeJ mice. In this proposal, we aim to elucidate this genetic basis of the sensitivity of the musculo-skeleton to the loss of appropriate mechanical signals by identifying the quantitative trait loci (QTL) responsible for the difference exhibited between these two strains of inbred mice. The identification of QTL (and ultimately the responsible genes) may be used as both a critical diagnostic sensor for the identification of astronauts that are in greatest need of pharmacologic and/or biomechanical countermeasures in space and as a discovery tool of novel drug targets for the preventing the loss of musculo-skeletal tissue. - to reveal the chromosomal locations (genes) that are associated with a high and low sensitivity of bone and muscle to mechanical disuse (tail suspension) by crossing a mechanically responsive (BALB/cByJ) and a mechanically unresponsive strain (C3H/HeJ), thus determing at the genetic level why only one of the two mouse strains responds to the loss of physical factors - to study whether the tight association between muscle and bone will cause QTLs for muscle to overlap with QTLs for bone. - to develop a genetic diagnostic sensor to identify individuals who at highest risk of loosing bone and muscle due to space flight, bedrest, or aging. - to investigate three-dimensional changes in trabecular bone architecture induced by mechanical disuse which may provide the basis for future finite element (FE) models to study the mechanical mechanisms of bone loss. - to probe the influence of genetics on the recovery of bone loss upon a period of unloading. - to serve as a discovery tool for novel drug targets that will prevent the loss of musculo-skeletal tissue. The very large variability in bone loss between astronauts (i.e., some lose a lot, some lose almost nothing), in the absence of large variations in environmental factors (e.g., diet or exercise), suggest a strong genetic influence on bone's susceptibility to the loss of functional weightbearing. The principal aim of this proposal to elucidate the genetic basis of the sensitivity of the musculo-skeleton to the loss of appropriate mechanical signals may lead to the design of a diagnostic sensor for the identification of astronauts that are in greatest need of pharmacologic and/or biomechanical countermeasures in space, as well as to the discovery of novel drug targets that will allow for safe and efficacious countermeasures. This proposal addresses two questions that are central to the Critical Path: Question 1g: What are the important predictors for estimating site-specific bone loss and fracture risk during hypogravity exposure, including, but not limited to ethnicity, gender, genetics, age, baseline bone density and geometry, nutritional status, fitness level and prior microgravity exposure? The focus of this grant is to investigate how genetics modulates bone quantity and micro-architecture and, more specifically, to identify the specific chromosomal regions that predispose individuals to the loss of muskulo-skeletal tissue in space. Knowledge of these chromosomal regions (QTL) may be used as a diagnostic tool before space travel (and the loss of tissue) and for the design of individually tailored treatment regimes. Further, as described below and in last year's report, we are also investigating how gender, baseline density and geometry affect site-specific bone loss. Question 1k: Are hypogravity-induced changes in bone density, geometry, and architecture reversible upon encountering partial gravity exposure, or on return to full gravity (1-G)? Mice that are subjected to hindlimb suspension in this study are allowed, upon completion of 3wk of unloading, to resume normal cage activities and any restoration in bone morphology and architecture is assessed by in vivo micro-computed tomography.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Skeletal pathologies involving bone loss, such as osteoporosis, are having a catastrophic impact upon the quality of life of the affected population and upon the cost of North American health care. In the United States alone, it is estimated that more than 44 million women and men (that is 55% of all people older than fifty years) are inflicted with osteoporosis or will be at high risk of developing it, currently costing the health care system more than $20 billion annually. Data derived from these studies will not only be relevant to spaceflight but be equally important to the prevention and treatment of bone loss associated with bedrest, musculo-skeletal diseases, and decreased levels of physical activity in our aging population.
	Over the past 12 months, large strides have been made towards completing the goals of this grant and morphological data from bone and muscle of first generation (F1) and second generation (F2) BALBxC3H crosses that were unloaded (and reloaded) for 3wk are presented. Specifically, we tested the hypotheses in this report that (1) the interaction between genetics and gender determines the magnitude and distribution of site-specific bone loss and that (2) genetically determined peak bone mass and micro-architecture can influence the rates of bone loss when the skeleton is exposed to unloading. METHODS Female and Male Mice of First Generation F1-Cross Sixteen-week-old male (M) and female (F) F1 crossbred (BALBxC3H) mice were randomly assigned to baseline (n=7M, 8F), age-matched control (n=10M, 10F), and hindlimb unloading (n=10M, 12F) groups. After 21d, the proximal tibia and distal femur were scanned using high-resolution (12um) micro-computed tomography (Fig. 1). Morphometric parameters including bone volume fraction (BV/TV), connectivity density (Conn.D), and trabecular number (Tb.N) and thickness (Tb.Th) were determined. Female Mice of Second Generation F2-Cross Identical to the experimental design above, F1 mice were crossed and a genetically heterogeneous BALBxC3H (F2) mouse population (n=250) was hind limb unloaded at 4mo of age for 3wk. A subset of these mice was subjected to a 3wk recovery phase upon completion of the unloading protocol. In-vivo micro-computed tomography (10.5?m resolution) quantified longitudinal changes in trabecular bone morphology of the distal femur as well as cross-sectional muscle and cortical bone geometry at the middiaphysial femur (20?m resolution). All animals were scanned at baseline as well as after unloading and reloading. Several subsets of mice were used to document bone's temporal response in greater detail (scanned every week) and to address the potential influence of radiation on bone loss (scanned at lower resolutions and radiation levels). RESULTS Female and Male Mice of First Generation F1-Cross At the start of the experiment, body masses were similar across all groups within a given gender. While body mass of male and female controls increased during the experiment (3% each, $p<0.05$ each), body mass in disuse males and females decreased (13%, 10%; $p<0.01$ each). Gender differences between age-matched control mice were associated with site- and bone-specific differences in

Task Progress:	trabecular bone morphology. Comparisons between baseline and age-matched controls within each gender revealed no differences in bone morphology in either bone, suggesting that bone did not accrue during the 21-day experiment. The effects of unloading on trabecular bone were seen in both males and females, but the magnitude of the effect varied according to anatomic site within a bone, across bones, and across gender. Trabecular bone volume fraction (BV/TV) was significantly reduced by disuse in the metaphysis (-30%, $p<0.01$), but was largely unaffected in the epiphysis. Similarly, disuse was associated with significantly lower cortical bone area in the metaphyseal cortex (-14%, $p<0.01$) but diaphyseal cortical bone area was not altered. Unloading in male mice significantly decreased the number of trabeculae in both the metaphysis (but not the epiphysis) of the proximal tibia and distal femur while trabecular number in female mice was altered by the loss of functional weight bearing in both the epiphysis and metaphysis of the femur (but not the tibia). These data strongly indicate that the effects of simulated weightlessness vary across bone tissue types, anatomical sites within a given bone, and between weight bearing bones. Female Mice of Second Generation F2-Cross Longitudinal in vivo micro-computed tomography allowed the quantification of altered bone morphology and architecture over time in an individual animal. Over the 3wk disuse period, bone volume fraction (BV/TV) in the distal femur diminished on average by 55%±16% ($p<0.0001$). Supporting the genetic influence on this large variability in bone loss, data were normally distributed with losses ranging from 17%-89%. The average rate of bone loss was similar between the first and second week (20% and 21%) but increased to 29% during the third week. Various exposure levels to cumulative radiation did not affect the magnitude of bone loss. Trabecular bone architecture also deteriorated dramatically with unloading; connectivity density decreased by 54% ($p<0.0001$), trabecular number by 22% ($p<0.0001$), and trabecular thickness by 17% ($p<0.0001$). Changes in bone volume fraction correlated ($R2=0.6$) with changes in trabecular number, connectivity density and trabecular thickness. Baseline bone volume fraction was a moderately good, but inverse, predictor of the magnitude of bone loss ($R2=0.4$). The effect of unloading was less evident in cortical bone where bone was resorbed primarily from the endocortical surface (-10%, $p<0.0001$). Removal of the functional environment from the hind limbs also affected the muscle groups surrounding the diaphyseal femur. Averaged across all mice, cross-sectional muscle area diminished by 15% (range: 0-27%) after 3wk of unloading. Muscle volume at baseline influenced the magnitude of muscle loss during disuse ($R2=0.2$). Baseline muscle area was correlated with baseline trabecular and cortical bone volume fraction ($R2=0.3$-0.5) but the magnitude of muscle loss was neither associated with cortical nor trabecular bone loss. The 3wk recovery period during, which all animals were re-introduced to habitual cage-activities, stopped or reversed the loss of muscle tissue experienced during disuse (range 0-110%). The degree of muscle loss correlated well ($R2=0.7$) with the degree of muscle recovery (i.e., mice displaying the greatest amount of tissue loss also regained the greatest amount). In great contrast to muscle, only 50% of all mice gained trabecular bone volume during the recovery period with the other 50% continuing to shed bone volume. DISCUSSION Consistent with data from astronauts showing very large individual variations in the amount of bone loss upon return from space, these microCT data from F1 (genetically homogeneous) and F2 (genetically heterogeneous) mice link the large variations in bone and muscle loss to differences in gender, genetics, bone type, and anatomical location. Also similar to human space flight data, disuse preferentially targeted trabecular sites within the femur, yet the ratio of trabecular to cortical bone loss was greater than observed during space missions. Gender's influence on site-specific bone morphology and its susceptibility to disuse osteopenia was investigated within the femur and tibia of adult male and female F1 mice. While gender effects on bone quantity and morphology have been described long ago, our data suggest that this gender influence may vary across anatomic sites, not only within a given bone but also across different bones. Not surprisingly, the effects of unloading on bone quantity also varied across gender and anatomic site, both within and across bones. Taken together, these results suggest that gender modulates the site-specific sensitivity of the skeleton to altered mechanical loading environments. Mice subjected to different levels of in vivo microCT induced radiation indicated that this methodology can be safely used to describe the longitudinal changes in bone morphology and architecture during simulated weightlessness. The moderately high correlations between baseline bone morphology and the magnitude of bone loss suggest that some genes may have simultaneous, if not inverse, influences on both traits. The disassociation between changes in muscle and bone volume during disuse and recovery from disuse further suggests that mechano-sensitivity of muscle and bone is not modulated by the same set of genes. Extrapolated to humans, these data indicate that people with lower peak bone mineral density may suffer from higher rates of bone but not muscle loss during disuse conditions. In summary, the strong influence of genetics was evident in baseline morphology as well as in the skeletal response to disuse. Currently, we are analyzing genomic DNA extracted from these genetically distinct mice and relate differences in genetic markers to differences in disuse-induced changes in bone and muscle morphology to determine the quantitative trait loci (QTL) that regulate the susceptibility of the musculo-skeleton to unloading. The identification of QTL (and ultimately the responsible genes) may be used as both a critical diagnostic sensor for the identification of astronauts that are in greatest need of pharmacologic and/or biomechanical countermeasures in space and for the selection of treatment tailored to the genetic make-up of an individual. Critical Path Question 1g: What are the important predictors for estimating site-specific bone loss and fracture risk during hypogravity exposure, including, but not limited to ethnicity, gender, genetics, age, baseline bone density and geometry, nutritional status, fitness level and prior microgravity exposure? Data described in this report infer that the sensitivity of a given anatomical location to the loss of functional weight bearing is heavily influenced by genetics, gender, and baseline morphology and geometry. The moderately high (but inverse) association between baseline bone morphology and the magnitude of bone loss emphasizes that those individuals that already have lower than normal levels of bone mass may be at greatest risk to lose large amounts of bone during space-flight. Further, the strong site-specificity by which genetics modulates bone loss will limit our ability to categorize entire skeletons into varying degrees of responsivity based upon their genotype. Hence, the optimal site of detection for spaceflight induced osteoporosis may depend on an individual's genotype and gender. Our current efforts to define those genes controlling this relation will ultimately allow the identification of those at greatest risk to incur substantial losses in bone and muscle tissue. Critical Path Question 1k: Are hypogravity-induced changes in bone density, geometry, and architecture reversible upon encountering partial gravity exposure, or on return to full gravity (1-G)? Similar to previous studies, our data indicate a limited potential for a bone degraded by disuse ostepeonia to recover its morphology (and strength). Perhaps more importantly, this study also suggests a strong genetic basis for this (partial) recovery and the genetic allele-typing that we are currently performing will be able to point towards the genes that are associated with this differential ability of individuals to regain some of their bone tissue.
Bibliography Type:	Description: (Last Updated: 01/16/2018)

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